

ROGERS CITY, MI, USA

WMO: 727417

Lat: 45.407N

Lon: 83.813W

Elev: 204

StdP: 98.90

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.9	-21.6	0.6	-17.1	-19.5	0.7	-14.9	10.1	-5.9	9.1	-5.6	3.5	260	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.0	20.7	27.9	19.9	26.3	19.3	22.5	27.3	21.5	25.8	20.6	24.5	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	16.1	25.2	20.1	15.1	24.0	19.1	14.2	23.0	67.3	27.3	63.4	26.1	60.3	24.5	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.1	7.3	-21.2	33.4	3.0	1.4	-23.4	34.4	-25.1	35.2	-26.8	36.0	-29.0	37.0		
(5)			-21.5	24.5	2.9	0.8	-23.6	25.1	-25.4	25.6	-27.0	26.1	-29.1	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-5.9	-5.8	-1.4	4.7	11.5	16.6	20.1	19.3	15.6	9.4	2.8	-2.5
	DBStd	10.33	5.49	5.43	5.81	4.73	4.84	3.92	3.20	3.06	4.36	4.74	4.76	4.57
	HDD10.0	2183	492	442	356	171	38	1	0	0	4	71	221	388
	HDD18.3	4287	751	676	611	409	222	78	19	25	106	280	466	646
	CDD10.0	1125	0	0	4	12	83	200	312	287	171	52	5	0
	CDD18.3	189	0	0	1	1	8	28	72	53	23	3	0	0
	CDH23.3	1269	0	0	7	7	88	205	487	304	149	23	0	0
	CDH26.7	319	0	0	2	1	27	58	121	68	40	3	0	0
Wind	WSAvg	3.3	3.6	3.9	3.7	3.7	3.1	2.7	2.7	2.8	3.0	3.5	3.6	3.7
Precipitation	PrecAvg	762	54	39	40	59	73	78	83	86	80	65	60	51
	PrecMax	875	98	78	124	92	169	155	157	179	161	156	108	126
	PrecMin	600	23	13	3	16	19	10	34	34	32	13	23	13
	PrecStd	77	23	20	26	23	34	36	35	41	32	32	24	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	9.9	21.5	23.1	30.1	31.8	32.3	31.6	31.0	26.4	19.3	11.4
		MCWB	5.8	7.1	14.1	13.7	18.6	21.3	22.2	21.5	21.1	17.6	12.7	9.4
	2%	DB	5.1	5.1	12.7	18.8	25.7	27.9	29.7	28.3	27.4	22.3	14.5	7.2
		MCWB	3.3	2.9	8.6	13.0	17.0	20.1	21.3	20.2	19.8	17.3	10.3	5.4
	5%	DB	2.8	2.8	9.0	15.0	22.5	25.9	27.7	26.4	24.7	19.1	12.0	5.0
		MCWB	1.6	1.0	5.3	9.7	15.6	18.8	20.1	19.8	18.9	15.4	8.9	3.5
	10%	DB	1.5	1.2	6.2	12.2	19.9	23.0	26.1	24.9	22.6	16.9	9.8	3.3
		MCWB	0.6	-0.1	3.2	7.2	14.4	17.3	19.4	19.2	18.0	13.9	7.5	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	6.8	15.1	15.4	20.8	23.0	24.0	23.0	23.0	20.4	14.1	9.9
		MCDB	7.5	9.2	21.4	20.3	26.9	29.0	29.4	27.2	28.3	24.3	17.9	10.5
	2%	WB	3.5	3.3	9.4	12.7	18.9	21.2	22.6	21.9	21.3	18.0	11.1	5.8
		MCDB	4.8	5.0	12.6	18.8	23.1	26.2	27.4	26.5	25.1	20.3	14.0	6.8
	5%	WB	1.8	1.3	5.6	10.0	16.9	19.8	21.5	21.0	20.1	15.9	9.2	3.7
		MCDB	2.7	2.5	8.5	14.3	21.6	23.9	25.8	24.5	23.6	18.8	11.6	4.7
	10%	WB	0.6	0.0	3.3	7.7	14.9	18.6	20.5	20.2	18.8	13.9	7.6	2.3
		MCDB	1.5	1.0	5.7	11.4	19.4	22.8	24.6	23.6	22.2	16.6	9.3	3.4
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	8.1	9.0	11.0	10.4	9.9	9.3	9.6	7.9	6.5	5.3
		MCDBR	6.9	8.2	12.3	14.6	15.0	12.9	12.0	11.5	12.1	12.2	10.5	7.5
		MCWBR	5.9	6.4	8.2	9.0	8.2	6.9	6.0	5.7	6.3	7.8	7.4	6.5
	5% WB	MCDBR	6.6	7.8	11.5	14.0	13.6	11.3	10.3	9.6	10.4	11.1	9.7	6.9
		MCWBR	5.9	6.7	8.3	9.2	8.6	7.0	5.9	5.4	6.4	8.1	7.6	6.4
Clear-Sky Solar Irradiance	taub	0.294	0.295	0.313	0.353	0.372	0.401	0.399	0.387	0.365	0.335	0.305	0.281	
	taud	2.360	2.342	2.409	2.354	2.381	2.331	2.354	2.386	2.439	2.500	2.525	2.466	
	Ebn at Noon	813	888	921	907	896	868	864	863	854	835	806	800	
	Edh at Noon	75	92	100	116	117	124	119	111	97	80	65	62	
All-Sky Solar Radiation	RadAvg	1.20	2.17	3.67	4.67	5.59	6.08	6.14	5.21	4.02	2.38	1.27	0.86	
	RadStd	0.12	0.21	0.35	0.45	0.51	0.46	0.23	0.28	0.36	0.26	0.18	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page